

Day 1 — AI Fundamentals & Prompting for TPMs

Activity packet for facilitators and participant triads. This file is the single source of truth for the day's small-group work, character cases, and deliverable templates.

Running case study — FieldPulse

Every day of Week 1 uses the same fictional product to keep momentum.

FieldPulse is a mobile-first field-service dispatch SaaS sold to HVAC, plumbing, and electrical shops with 50–200 technicians. It competes with ServiceTitan, Housecall Pro, and in-house tooling. It is mid-sized (Series B), 400 customers, \$40M ARR. The academy's "TPM role" sits in the Dispatcher Workflow squad, which owns the dispatcher's primary web app and the tech's mobile app.

What the cohort will be given as ambient context

- One-page product overview (handed out at start of Day 1)
- Three anonymized dispatcher interview transcripts (used starting Day 3)
- A pack of 14 support tickets (used on Day 4)
- A screen-time analytics export (Day 4)
- A composite customer quote from a prior PR (Day 2)

Pre-flight: print or digitally share these handouts before Day 1 morning break.

Triad formation

Form triads by 10:30 on Day 1. Composition rules:

1. Mix experience levels (at least one person with 3+ years in an engineering-adjacent role per triad where possible)
2. Mix domain backgrounds
3. No two people from the same home team if cohort is intra-company

Triads persist through the **end of Week 3** (they carry the mini-capstone). Make the formation stick.

Activity 1 — "Same Question, Three Models"

Format: Triad • 35 min • Block 1

Purpose

Surface that (a) different AI tools give materially different answers to the same question, and (b) "it sounded confident" is not evidence of correctness.

Setup

Each triad picks **one** question from the list below. Each triad member runs it on a different tool (ChatGPT, Claude, Gemini/Copilot). If only one tool is available to the whole cohort, vary **phrasing** across the three triad members instead.

The question bank (pick one)

1. What's the largest regulatory risk a dispatcher-facing SaaS should plan for when expanding into HVAC in California?
2. Give me the five most frequently stated frustrations of HVAC dispatchers, with sources.
3. What are the top 3 dispatch-tool competitors in the U.S. small-mid market, and their differentiators?
4. Summarize the dispatcher-to-tech handoff process in a 50-person HVAC company.
5. What KPIs do dispatcher teams actually report to owners?
6. Walk me through how a dispatcher typically reacts when a technician calls out sick mid-shift.
7. List integrations a small HVAC shop likely already uses.
8. How do most dispatch apps handle offline/intermittent connectivity today?
9. Who is Maria, a fictional HVAC dispatcher? (watch for hallucinated specifics)
10. What academic research supports mobile-first field service tools?

Steps

1. **Pick (5 min)**. Triad selects one question from the bank above.
2. **Run (10 min)**. Each triad member runs the same question on a different tool.
3. **Score (15 min)**. Compare outputs across usefulness, verifiability, and risk-if-wrong using the rubric below.
4. **Prepare readout (5 min)**. Draft a one-line trust statement.

Scoring rubric (each tool gets three 1–5 scores)

DIMENSION	1	5
Usefulness	Generic/off-topic	Specific to domain & role
Verifiability	Claims can't be checked	Claims traceable, named sources
Risk if wrong	Low (opinion, easy to fix)	High (would mislead scoping or strategy)

Deliverable

A scored comparison across three tools plus a one-line readout: "We would trust [Tool X] for [kind of question]. We would never trust any of them for [kind of question]."

Facilitation cues

- If all three tools agreed, ask "did they agree *correctly*? How do you know?" — surfaces the comfortable-but-wrong trap.
- If a tool fabricated a citation, celebrate it as a finding. Take a screenshot for the week's reference.

Activity 2 — Rewrite This Prompt

Format: Triad • 40 min • Block 2

Purpose

Convert the Day 1 teaching (RCCF) into reflexive behavior. Participants handle three weak prompts pulled from the FieldPulse backlog and rewrite them.

Setup

Distribute the three weak prompts below to each triad. Confirm each triad has access to one AI assistant and the FieldPulse handouts from the morning.

The three weak prompts

1. "Write me an epic for improving the reconcile workflow."
2. "Summarize all the dispatcher interview feedback."
3. "Give me a competitive comparison chart."

Steps

1. Identify what's missing: Role, Context, Constraints, Format
2. Add specifics from the FieldPulse handouts
3. Add a constraint that limits hallucination (e.g., "cite evidence," "flag assumptions")
4. Specify exact output format (markdown table, 3-bullet list, etc.)

Deliverable — Prompt Pattern Library (seed)

Each triad publishes a shared document (any format) with this schema:

```
## Pattern: <short name>

**When to use:** <situation>
**Template:**
```

Role: ... Context: ... Constraints: ... Format: ...

```
**Why it works:** <1-2 sentences>
**Caveats:** <where it still fails>
```

Minimum four entries by end of Day 1. Entries earned in Activities 2, 3, and 4.

Facilitation cues

- Walk the room while triads rewrite. The most common gap is under-specified Constraints — coach by asking, not telling.
- If a triad's rewrite still gets a vague output, that's a teaching moment for the cohort: weak prompts can survive one improvement and still fail.

Activity 3 — The Three Hats

Format: Triad • 40 min • Block 3

Purpose

Use AI as a rapid multi-perspective critic, and practice asking a single question from each perspective.

Setup

Each triad has a working AI assistant and the FieldPulse one-paragraph problem brief below. The brief is the shared input for all three hats.

The FieldPulse one-paragraph problem brief

Dispatchers at mid-sized HVAC shops spend roughly 45 minutes after their shift reconciling paper tickets, truck stock, and tech timecards. We believe a mobile-first guided reconcile flow could cut this to under 10 minutes. We are considering building this in Q3.

Steps

1. **Engineer hat (10 min)**. Prompt: "Read this as a skeptical engineer scoping the work. What three questions will you ask first?"
2. **Blocker-stakeholder hat (10 min)**. Prompt: "Read this as the operations VP who most often blocks dispatcher-app launches here (she's worried about training load). What objection worries you most?"
3. **Target-customer hat (10 min)**. Prompt: "Read this as Maria, a 12-year HVAC dispatcher at a 90-tech shop. Where does this not ring true?"
4. **Synthesize (10 min)**. Pick one question per hat that is most uncomfortable; pick one that becomes the Day 3 research target.

Deliverable

For each hat: the single most uncomfortable question surfaced. One of those three questions becomes your **Day 3 research target** (the one you cannot currently answer).

Facilitation cues

- Watch for "the AI was nice to us" outputs — push the triad to re-prompt with sharper persona constraints.
- The discomfort is the point. If all three hats produce comfortable questions, the prompting is too soft.

Activity 4 — Prompt Pattern Library (Publish)

Format: Triad • 45 min • Block 4 + readouts

Purpose

Ship a durable artifact. The library is referenced every remaining day of Week 1 and beyond.

Setup

Each triad has been collecting candidate prompts through Activities 2 and 3. They now formalize a shared document — any tool the triad will reliably use (markdown file, shared doc, wiki page).

Steps

1. **Inventory (10 min)**. Pull every candidate prompt the triad has used today.
2. **Cull (15 min)**. Keep the ones with a clear "when to use" and a known failure mode.
3. **Format (15 min)**. Convert each to the schema below. Minimum four entries.
4. **Star (5 min)**. Pick the prompt the triad would actually reuse on Monday morning.

Required entries (minimum)

- ☐ One Research prompt
- ☐ One Summarization-with-evidence prompt
- ☐ One Critique-hat prompt
- ☐ One "Where could you be wrong?" prompt

Optional extra credit

- ☐ One prompt for converting tickets → themed frustrations (will be needed on Day 4)
- ☐ One prompt for generating candidate JTBDs from a persona (Day 3)

Deliverable

A shared Prompt Pattern Library with at least four entries, each tagged with when-to-use, the template, why it works, and known caveats. One entry is starred as the triad's go-to.

Readout structure (60 seconds per triad)

"Our starred prompt is . We tested it on . The reason we'd reuse it is . Its known failure mode is ."

Facilitation cues

- Post all triad libraries to a shared space by end of Day 1. Every triad should be able to borrow another's prompts starting Day 2.
- Triads that finish early often have shallow entries. Push them: "What's the smallest change that would break this prompt?"

End-of-day checkpoint

Each triad leaves the day with:

- ☒ Formed triad with declared norms (meet-time, tool choice, note-keeper role)
- ☒ At least one person per triad who owns the Pattern Library file
- ☒ A Day 3 research target question selected from Activity 3
- ☒ A pre-Day-2 commitment: one product idea they want to PR/FAQ tomorrow

Facilitator reflection prompts (end of day)

- Did any triad over-rely on AI without critique? What would I model differently tomorrow?
- Which failure mode (hallucination, starvation, sycophancy, anchoring) did the room most struggle to see?
- Who in the room was quieter than I'd like? How can I engineer tomorrow's triad interactions to pull them in?